

BENSON LIN

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EDUCATION

Worcester Polytechnic Institute

Master of Science in Computer Science

May 2026

Worcester, MA

Worcester Polytechnic Institute

Bachelor of Science in Computer Science (GPA: 3.56/4.0)

May 2026

Worcester, MA

SKILLS

Languages : Python, Java, TypeScript, JavaScript, SQL

Backend : FastAPI, Spring Boot, REST APIs, SQLAlchemy, Spring Data JPA, Hibernate, WebSockets

Databases & Infrastructure : PostgreSQL, Docker, Flyway, Maven, GitHub Actions

Testing & Frontend : JUnit 5, Testcontainers, Vitest, React, Next.js, Tailwind CSS, Git

EXPERIENCE

Microsoft Intune (Sponsored MQP)

Aug. 2025 – Mar. 2026

Capstone Software Engineer

Remote

- Built a Python telemetry pipeline that captures real-time Win32 UI events and converts them into semantic workflow graphs to surface user friction and UI latency.
- Combined heuristic detectors with a Hidden Markov Model sequence detector (Baum-Welch training, Viterbi decoding) into a single workflow-level system, reaching **90.1% F1** (92.8% precision, 87.6% recall) on a human-labeled dataset.
- Cut false positives by **65%** through threshold tuning and failure analysis of dead-click inflation and application-error blind spots.

SerenDPT (Venice Project Center IQP)

Mar. 2025 – Jun. 2025

Software Developer

Venice, Italy

- Wrote Python and Node.js scripts to collect, clean, and validate infrastructure and rental data across **126 historic islands** for a Venice Project Center study.

PROJECTS

Herd | herd-wpi.vercel.app

Aug. 2026 – Present

- Built and deployed a full-stack campus feedback platform (FastAPI, PostgreSQL, Next.js, TypeScript) hosted live on Railway and Vercel with an algorithmically ranked feed.
- Engineered backend anti-abuse systems including multi-tier rate limiting, Cloudflare Turnstile verification, and database row-level locking to prevent race conditions, verified by 320+ automated tests in pytest.

DropForge | dropforge-roan.vercel.app

Jun. 2026 – Aug. 2026

- Built a limited-drop reservation and checkout backend (Spring Boot, PostgreSQL, Flyway) with idempotent reservation creation and an atomic conditional inventory update that prevents overselling under concurrent requests; pessimistic row locks serialize checkout and cancellation.
- Wrote **80 automated tests** (22 unit, 58 integration) with JUnit 5 and Testcontainers against real PostgreSQL, run in GitHub Actions CI, including failure-injection tests that confirm a mid-transaction error rolls back the inventory change.

Vigil | getvigilapp.vercel.app

May 2026 – Jul. 2026

- Built a telemetry ingestion API (FastAPI, async SQLAlchemy, PostgreSQL) utilizing partial unique indexes to prevent concurrent requests from generating duplicate incidents.
- Consolidated the service-health dashboard's three queries into a single round-trip using a ROW_NUMBER window function for each service's latest p99 latency, error rate, and open-incident severity.